

MAINTENANCE	MA-1	MA-1	SYSTEM MAINTENANCE POLICY AND PROCEDURES	The organization:
MAINTENANCE	MA-1	MA-1a	SYSTEM MAINTENANCE POLICY AND PROCEDURES	Develops, documents, and disseminates to [Assignment: organization-defined personnel or roles]:

MAINTENANCE	MA-1	MA-1a.1	SYSTEM MAINTENANCE POLICY AND PROCEDURES	A system maintenance policy that addresses purpose, scope, roles, responsibilities, management commitment, coordination among organizational entities, and compliance; and
MAINTENANCE	MA-1	MA-1a.2	SYSTEM MAINTENANCE POLICY AND PROCEDURES	Procedures to facilitate the implementation of the system maintenance policy and associated system maintenance controls; and

MAINTENANCE	MA-1	MA-1b	SYSTEM MAINTENANCE POLICY AND PROCEDURES	Reviews and updates the current:
MAINTENANCE	MA-1	MA-1b.1	SYSTEM MAINTENANCE POLICY AND PROCEDURES	System maintenance policy [Assignment: organization-defined frequency]; and

MAINTENANCE	MA-1	MA-1b.2	SYSTEM MAINTENANCE POLICY AND PROCEDURES	System maintenance procedures [Assignment: organization-defined frequency].
MAINTENANCE	MA-2	MA-2	CONTROLLED MAINTENANCE	The organization:

MAINTENANCE	MA-2	MA-2a	CONTROLLED MAINTENANCE	Schedules, performs, documents, and reviews records of maintenance and repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements;
MAINTENANCE	MA-2	MA-2b	CONTROLLED MAINTENANCE	Approves and monitors all maintenance activities, whether performed on site or remotely and whether the equipment is serviced on site or removed to another location;

MAINTENANCE	MA-2	MA-2c	CONTROLLED MAINTENANCE	Requires that [Assignment: organization-defined personnel or roles] explicitly approve the removal of the information system or system components from organizational facilities for off-site maintenance or repairs;
MAINTENANCE	MA-2	MA-2d	CONTROLLED MAINTENANCE	Sanitizes equipment to remove all information from associated media prior to removal from organizational facilities for off-site maintenance or repairs;

MAINTENANCE	MA-2	MA-2e	CONTROLLED MAINTENANCE	Checks all potentially impacted security controls to verify that the controls are still functioning properly following maintenance or repair actions; and
MAINTENANCE	MA-2	MA-2f	CONTROLLED MAINTENANCE	Includes [Assignment: organization-defined maintenance-related information] in organizational maintenance records.

MAINTENANCE	MA-2 (1)	MA-2 (1)	RECORD CONTENT	[Withdrawn: Incorporated into MA-2].
MAINTENANCE	MA-2 (2)	MA-2 (2)	AUTOMATED MAINTENANCE ACTIVITIES	The organization:
MAINTENANCE	MA-2 (2)	MA-2 (2)(a)	AUTOMATED MAINTENANCE ACTIVITIES	Employs automated mechanisms to schedule, conduct, and document maintenance and repairs; and
MAINTENANCE	MA-2 (2)	MA-2 (2)(b)	AUTOMATED MAINTENANCE ACTIVITIES	Produces up-to date, accurate, and complete records of all maintenance and repair actions requested, scheduled, in process, and completed.

MAINTENANCE	MA-3	MA-3	MAINTENANCE TOOLS	The organization approves, controls, and monitors information system maintenance tools.
MAINTENANCE	MA-3 (1)	MA-3 (1)	INSPECT TOOLS	The organization inspects the maintenance tools carried into a facility by maintenance personnel for improper or unauthorized modifications.

MAINTENANCE	MA-3 (2)	MA-3 (2)	INSPECT MEDIA	The organization checks media containing diagnostic and test programs for malicious code before the media are used in the information system.
MAINTENANCE	MA-3 (3)	MA-3 (3)	PREVENT UNAUTHORIZED REMOVAL	The organization prevents the unauthorized removal of maintenance equipment containing organizational information by:

MAINTENANCE	MA-3 (3)	MA-3 (3)(a)	PREVENT UNAUTHORIZED REMOVAL	Verifying that there is no organizational information contained on the equipment;
MAINTENANCE	MA-3 (3)	MA-3 (3)(b)	PREVENT UNAUTHORIZED REMOVAL	Sanitizing or destroying the equipment;

MAINTENANCE	MA-3 (3)	MA-3 (3)(c)	PREVENT UNAUTHORIZED REMOVAL	Retaining the equipment within the facility; or
MAINTENANCE	MA-3 (3)	MA-3 (3)(d)	PREVENT UNAUTHORIZED REMOVAL	Obtaining an exemption from [Assignment: organization-defined personnel or roles] explicitly authorizing removal of the equipment from the facility.

MAINTENANCE	MA-3 (4)	MA-3 (4)	RESTRICTED TOOL USE	The information system restricts the use of maintenance tools to authorized personnel only.
MAINTENANCE	MA-4	MA-4	NONLOCAL MAINTENANCE	The organization:

MAINTENANCE	MA-4	MA-4a	NONLOCAL MAINTENANCE	Approves and monitors nonlocal maintenance and diagnostic activities;
MAINTENANCE	MA-4	MA-4b	NONLOCAL MAINTENANCE	Allows the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy and documented in the security plan for the information system;

MAINTENANCE	MA-4	MA-4c	NONLOCAL MAINTENANCE	Employs strong authenticators in the establishment of nonlocal maintenance and diagnostic sessions;
MAINTENANCE	MA-4	MA-4d	NONLOCAL MAINTENANCE	Maintains records for nonlocal maintenance and diagnostic activities; and

MAINTENANCE	MA-4	MA-4e	NONLOCAL MAINTENANCE	Terminates session and network connections when nonlocal maintenance is completed.
MAINTENANCE	MA-4 (1)	MA-4 (1)	AUDITING AND REVIEW	The organization:
MAINTENANCE	MA-4 (1)	MA-4 (1)(a)	AUDITING AND REVIEW	Audits nonlocal maintenance and diagnostic sessions [Assignment: organization-defined audit events]; and
MAINTENANCE	MA-4 (1)	MA-4 (1)(b)	AUDITING AND REVIEW	Reviews the records of the maintenance and diagnostic sessions.

MAINTENANCE	MA-4 (2)	MA-4 (2)	DOCUMENT NONLOCAL MAINTENANCE	The organization documents in the security plan for the information system, the policies and procedures for the establishment and use of nonlocal maintenance and diagnostic connections.
MAINTENANCE	MA-4 (3)	MA-4 (3)	COMPARABLE SECURITY / SANITIZATION	The organization:

MAINTENANCE	MA-4 (3)	MA-4 (3)(a)	COMPARABLE SECURITY / SANITIZATION	Requires that nonlocal maintenance and diagnostic services be performed from an information system that implements a security capability comparable to the capability implemented on the system being serviced; or
MAINTENANCE	MA-4 (3)	MA-4 (3)(b)	COMPARABLE SECURITY / SANITIZATION	Removes the component to be serviced from the information system prior to nonlocal maintenance or diagnostic services, sanitizes the component (with regard to organizational information) before removal from organizational facilities, and after the service is performed, inspects and sanitizes the component (with regard to potentially malicious software) before reconnecting the component to the information system.

MAINTENANCE	MA-4 (4)	MA-4 (4)	AUTHENTICATION / SEPARATION OF MAINTENANCE SESSIONS	The organization protects nonlocal maintenance sessions by:
MAINTENANCE	MA-4 (4)	MA-4 (4)(a)	AUTHENTICATION / SEPARATION OF MAINTENANCE SESSIONS	Employing [Assignment: organization-defined authenticators that are replay resistant]; and
MAINTENANCE	MA-4 (4)	MA-4 (4)(b)	AUTHENTICATION / SEPARATION OF MAINTENANCE SESSIONS	Separating the maintenance sessions from other network sessions with the information system by either:
MAINTENANCE	MA-4 (4)	MA-4 (4)(b)(1)	AUTHENTICATION / SEPARATION OF MAINTENANCE SESSIONS	Physically separated communications paths; or
MAINTENANCE	MA-4 (4)	MA-4 (4)(b)(2)	AUTHENTICATION / SEPARATION OF MAINTENANCE SESSIONS	Logically separated communications paths based upon encryption.
MAINTENANCE	MA-4 (5)	MA-4 (5)	APPROVALS AND NOTIFICATIONS	The organization:
MAINTENANCE	MA-4 (5)	MA-4 (5)(a)	APPROVALS AND NOTIFICATIONS	Requires the approval of each nonlocal maintenance session by [Assignment: organization-defined personnel or roles]; and
MAINTENANCE	MA-4 (5)	MA-4 (5)(b)	APPROVALS AND NOTIFICATIONS	Notifies [Assignment: organization-defined personnel or roles] of the date and time of planned nonlocal maintenance.

MAINTENANCE	MA-4 (6)	MA-4 (6)	CRYPTOGRAPHIC PROTECTION	The information system implements cryptographic mechanisms to protect the integrity and confidentiality of nonlocal maintenance and diagnostic communications.
MAINTENANCE	MA-4 (7)	MA-4 (7)	REMOTE DISCONNECT VERIFICATION	The information system implements remote disconnect verification at the termination of nonlocal maintenance and diagnostic sessions.

MAINTENANCE	MA-5	MA-5	MAINTENANCE PERSONNEL	The organization:
MAINTENANCE	MA-5	MA-5a	MAINTENANCE PERSONNEL	Establishes a process for maintenance personnel authorization and maintains a list of authorized maintenance organizations or personnel;

MAINTENANCE	MA-5	MA-5b	MAINTENANCE PERSONNEL	Ensures that non-escorted personnel performing maintenance on the information system have required access authorizations; and
MAINTENANCE	MA-5	MA-5c	MAINTENANCE PERSONNEL	Designates organizational personnel with required access authorizations and technical competence to supervise the maintenance activities of personnel who do not possess the required access authorizations.

MAINTENANCE	MA-5 (1)	MA-5 (1)	INDIVIDUALS WITHOUT APPROPRIATE ACCESS	The organization:
MAINTENANCE	MA-5 (1)	MA-5 (1)(a)	INDIVIDUALS WITHOUT APPROPRIATE ACCESS	Implements procedures for the use of maintenance personnel that lack appropriate security clearances or are not U.S. citizens, that include the following requirements:

MAINTENANCE	MA-5 (1)	MA-5 (1)(a)(1)	INDIVIDUALS WITHOUT APPROPRIATE ACCESS	Maintenance personnel who do not have needed access authorizations, clearances, or formal access approvals are escorted and supervised during the performance of maintenance and diagnostic activities on the information system by approved organizational personnel who are fully cleared, have appropriate access authorizations, and are technically qualified;
MAINTENANCE	MA-5 (1)	MA-5 (1)(a)(2)	INDIVIDUALS WITHOUT APPROPRIATE ACCESS	Prior to initiating maintenance or diagnostic activities by personnel who do not have needed access authorizations, clearances or formal access approvals, all volatile information storage components within the information system are sanitized and all nonvolatile storage media are removed or physically disconnected from the system and secured; and

MAINTENANCE	MA-5 (1)	MA-5 (1)(b)	INDIVIDUALS WITHOUT APPROPRIATE ACCESS	Develops and implements alternate security safeguards in the event an information system component cannot be sanitized, removed, or disconnected from the system.
MAINTENANCE	MA-5 (2)	MA-5 (2)	SECURITY CLEARANCES FOR CLASSIFIED SYSTEMS	The organization ensures that personnel performing maintenance and diagnostic activities on an information system processing, storing, or transmitting classified information possess security clearances and formal access approvals for at least the highest classification level and for all compartments of information on the system.
MAINTENANCE	MA-5 (3)	MA-5 (3)	CITIZENSHIP REQUIREMENTS FOR CLASSIFIED SYSTEMS	The organization ensures that personnel performing maintenance and diagnostic activities on an information system processing, storing, or transmitting classified information are U.S. citizens.
MAINTENANCE	MA-5 (4)	MA-5 (4)	FOREIGN NATIONALS	The organization ensures that:

MAINTENANCE	MA-5 (4)	MA-5 (4)(a)	FOREIGN NATIONALS	Cleared foreign nationals (i.e., foreign nationals with appropriate security clearances), are used to conduct maintenance and diagnostic activities on classified information systems only when the systems are jointly owned and operated by the United States and foreign allied governments, or owned and operated solely by foreign allied governments; and
MAINTENANCE	MA-5 (4)	MA-5 (4)(b)	FOREIGN NATIONALS	Approvals, consents, and detailed operational conditions regarding the use of foreign nationals to conduct maintenance and diagnostic activities on classified information systems are fully documented within Memoranda of Agreements.
MAINTENANCE	MA-5 (5)	MA-5 (5)	NONSYSTEM-RELATED MAINTENANCE	The organization ensures that non-escorted personnel performing maintenance activities not directly associated with the information system but in the physical proximity of the system, have required access authorizations.

MAINTENANCE	MA-6	MA-6	TIMELY MAINTENANCE	The organization obtains maintenance support and/or spare parts for [Assignment: organization-defined information system components] within [Assignment: organization-defined time period] of failure.
MAINTENANCE	MA-6 (1)	MA-6 (1)	PREVENTIVE MAINTENANCE	The organization performs preventive maintenance on [Assignment: organization-defined information system components] at [Assignment: organization-defined time intervals].
MAINTENANCE	MA-6 (2)	MA-6 (2)	PREDICTIVE MAINTENANCE	The organization performs predictive maintenance on [Assignment: organization-defined information system components] at [Assignment: organization-defined time intervals].
MAINTENANCE	MA-6 (3)	MA-6 (3)	AUTOMATED SUPPORT FOR PREDICTIVE MAINTENANCE	The organization employs automated mechanisms to transfer predictive maintenance data to a computerized maintenance management system.